

Math Camp 2026 — Day 3 Exercises

Probability · Visualization and Quarto

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Part I — Probability

1. Reading a table

Three hundred country-years, by regime type and compliance with an export-control regime.

	Complied	Did not	Total
Democracy	96	24	120
Autocracy	72	108	180
Total	168	132	300

- (a) Give $P(\text{Dem} \cap \text{Comp})$, $P(\text{Dem})$, $P(\text{Comp})$.
- (b) Give $P(\text{Comp} \mid \text{Dem})$ and $P(\text{Dem} \mid \text{Comp})$. Both use the cell 96 — state in one sentence each what question they answer.
- (c) Are regime type and compliance independent? Show the calculation.

2. Bayes

A port screening algorithm flags 90% of illicit containers and 5% of legitimate ones. Two percent of containers are illicit.

- (a) Compute $P(\text{illicit} \mid \text{flagged})$.
- (b) The test catches 90% of illicit containers. Explain in two sentences why your answer is so much lower.
- (c) Redo (a) if the algorithm is used only on a pre-screened subset that is 20% illicit. What follows about where to deploy such a tool?

3. Expectation and variance

X = number of veto players.

x	1	2	3	4
$p(x)$	0.3	0.4	0.2	0.1

- (a) Compute $E[X]$, $E[X^2]$, $\text{Var}(X)$.
- (b) Compare $E[X^2]$ with $(E[X])^2$. What is the gap, and why?
- (c) For $Y = 5X - 3$, give $E[Y]$ and $\text{Var}(Y)$ using the transformation rules.

4. Variance of sums, and correlation

Let $\text{Var}(X) = 4$, $\text{Var}(Y) = 9$, $\text{Cov}(X, Y) = -2$.

- (a) Compute $\text{Var}(X + Y)$, $\text{Var}(X - Y)$, and ρ .
- (b) Which of these could you compute knowing only $E[X]$ and $E[Y]$ — and which not? State the general principle in one sentence.
- (c) Now let $X \in \{-2, -1, 0, 1, 2\}$ with equal probability and $Y = X^2$. Show $\text{Cov}(X, Y) = 0$, then show X and Y are not independent.

5. Distributions and the CLT

- (a) Eight states each ratify independently with $p = 0.25$. Compute $P(\text{exactly two})$ and $P(\text{at least one})$, and give the R command for each.
- (b) A colleague's monthly protest counts have mean 2.4 and variance 9.1. Which Poisson assumption fails, and what does that do to standard errors?
- (c) Vote share is $N(48, 6^2)$. Find $P(X > 60)$ and the 95th percentile, with the R command for each.
- (d) With $\sigma = 12$ and $n = 36$, give $SE(\bar{X})$, and the n needed to halve it.
- (e) True or false, with one line of justification each:
 - i. The CLT implies X is approximately normal in the population.
 - ii. The sampling distribution of $\bar{X}$ is centred on μ .
 - iii. The usual SE is fine for 36 students drawn from three classrooms.

Part II — Computing

Use `analysis.rds` from yesterday. Write and run the code.

6. Inside or outside `aes()`

Predict what each line does *before* running it, then run all three.

```
ggplot(analysis, aes(gdp_pc, vote_share)) + geom_point(color = "region")
ggplot(analysis, aes(gdp_pc, vote_share, color = "steelblue")) + geom_point()
ggplot(analysis, aes(gdp_pc, vote_share, color = region)) + geom_point()
```

- (a) Which errors, which produces a wrong figure silently, and which is correct?
- (b) State the rule in one sentence.
- (c) Fix the first two so that one colours by region and the other makes every point steelblue.

7. Lines and panels

```
ggplot(analysis, aes(year, gdp_pc)) + geom_line()
```

- (a) Run it. Explain in one sentence why the result looks the way it does.
- (b) Fix it so there is one line per country.
- (c) Facet that by region. Then add `scales = "free_y"` and say what you gain and what you lose.

8. A pipeline into a plot

Starting from `analysis`, in a single chained statement:

- (a) Compute mean GDP per capita by region and year, and plot it as one coloured line per region.
- (b) Mark where `|>` stops and `+` begins, and say why the switch exists.
- (c) Add `scale_color_viridis_d()`. Give one reason beyond aesthetics.

9. Labelling and saving

Take the scatterplot of `vote_share` against `gdp_pc` on a log x axis.

- (a) Add a title, subtitle, axis labels with units, and a source caption. Apply `theme_minimal()`.
- (b) Save it as a PDF at 6.5×4.5 inches, and as a PNG at 300 dpi.
- (c) Save the same plot at 4×3 and at 10×7 inches. Open both. Explain the difference in apparent text size, and say which direction fixes labels that came out too small.

10. A Quarto document that will not render

Create `day3.qmd` with a YAML header, a setup chunk, one figure, and one sentence of prose reporting the number of observations via inline code.

- (a) Which chunk option runs the code and shows nothing? Which runs it and hides only the code?
- (b) Run `rm(list = ls())` and render. If you get `object 'analysis' not found`, say what caused it and where the fix goes.
- (c) Explain in one sentence why rendering in a clean session is a feature rather than an inconvenience.